Investigating the Association between Census Demographics and Hosting Capactiy

Load libraries & data

```
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.2.1     ✔ readr     2.1.4
✔ forcats   1.0.0     ✔ stringr   1.6.0
✔ ggplot2   4.0.3     ✔ tibble    3.3.1
✔ lubridate 1.9.2     ✔ tidyr     1.3.2
✔ purrr     1.2.2
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to
become errors
```

```
library(sfarrow)
library(tidycensus)
library(sf)
```

```
Linking to GEOS 3.10.2, GDAL 3.4.1, PROJ 8.2.1; sf_use_s2() is TRUE
```

```
library(here)
```

```
here() starts at /Users/zloo/MEDS/statistics
```

```
library(tidymodels)
```

```
── Attaching packages ──────────────────────────────────── tidymodels 1.5.0 ──
✔ broom        1.0.12     ✔ rsample      1.3.2
✔ dials        1.4.3      ✔ tailor       0.1.0
✔ infer        1.1.0      ✔ tune         2.1.0
✔ modeldata    1.5.1      ✔ workflows    1.3.0
✔ parsnip      1.5.0      ✔ workflowsets 1.1.1
✔ recipes      1.3.2      ✔ yardstick    1.4.0
── Conflicts ─────────────────────────────────────── tidymodels_conflicts() ──
✖ scales::discard() masks purrr::discard()
✖ dplyr::filter()   masks stats::filter()
✖ recipes::fixed()  masks stringr::fixed()
✖ dplyr::lag()      masks stats::lag()
✖ yardstick::spec() masks readr::spec()
✖ recipes::step()   masks stats::step()
```

```r
census <- read_csv('census_final_tidy.csv') |>
  janitor::clean_names() |>
  select(-geometry)
```

```
Rows: 200838 Columns: 8
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr (6): GEOID, NAME, variable, label, concept, geometry
dbl (2): estimate, moe

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
census_area <- st_read("~/../../capstone/electrigrid/data/census/census_tracts_2025/tl_202
  select(GEOID, ALAND) |> janitor::clean_names()
```

```
Reading layer `tl_2025_06_tract' from data source
  `/capstone/electrigrid/data/census/census_tracts_2025/tl_2025_06_tract.shp'
  using driver `ESRI Shapefile'

Warning in CPL_read_ogr(dsn, layer, query, as.character(options), quiet, : GDAL
Error 1: PROJ: proj_identify: /opt/anaconda3/share/proj/proj.db lacks
```

```
DATABASE.LAYOUT.VERSION.MAJOR / DATABASE.LAYOUT.VERSION.MINOR metadata. It
comes from another PROJ installation.

Simple feature collection with 9129 features and 13 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -124.482 ymin: 32.52951 xmax: -114.1312 ymax: 42.0095
Geodetic CRS:  GCS_North_American_1983
```

```
dac <- st_read_parquet("~/../../capstone/electrigrid/data/dis_adv_comm/calenviroscreen.par
  mutate(tract = as.character(tract)) %>%
  mutate(tract = str_pad(tract, width = 11, side = "left", pad = "0"))

dir <- here::here('~', '..', '..', 'capstone', 'electrigrid', 'data', 'census_hosting_capa

sce_census_map <- sfarrow::st_read_parquet(here(dir, "sce_census_map_new.parquet")) |>
    janitor::clean_names() |>
    select(geoid, zillow_tract_hh_count, starts_with('avg_der'), geometry)

pge_census_map <- sfarrow::st_read_parquet(here(dir, "pge_census_map_new.parquet")) |>
    janitor::clean_names() |>
    select(geoid, zillow_tract_hh_count, starts_with('avg_der'), geometry)

sdge_census_map <- sfarrow::st_read_parquet(here(dir, "sdge_census_map_new.parquet")) |>
    janitor::clean_names() |>
    select(geoid, zillow_tract_hh_count, starts_with('avg_der'), geometry)
```

Clean data

```
# combine all utilities
all_census_map <- bind_rows(sce_census_map, pge_census_map, sdge_census_map) |>
    # set minimum zillow count
    mutate(zillow_tract_hh_count = pmax(zillow_tract_hh_count, 1))
```

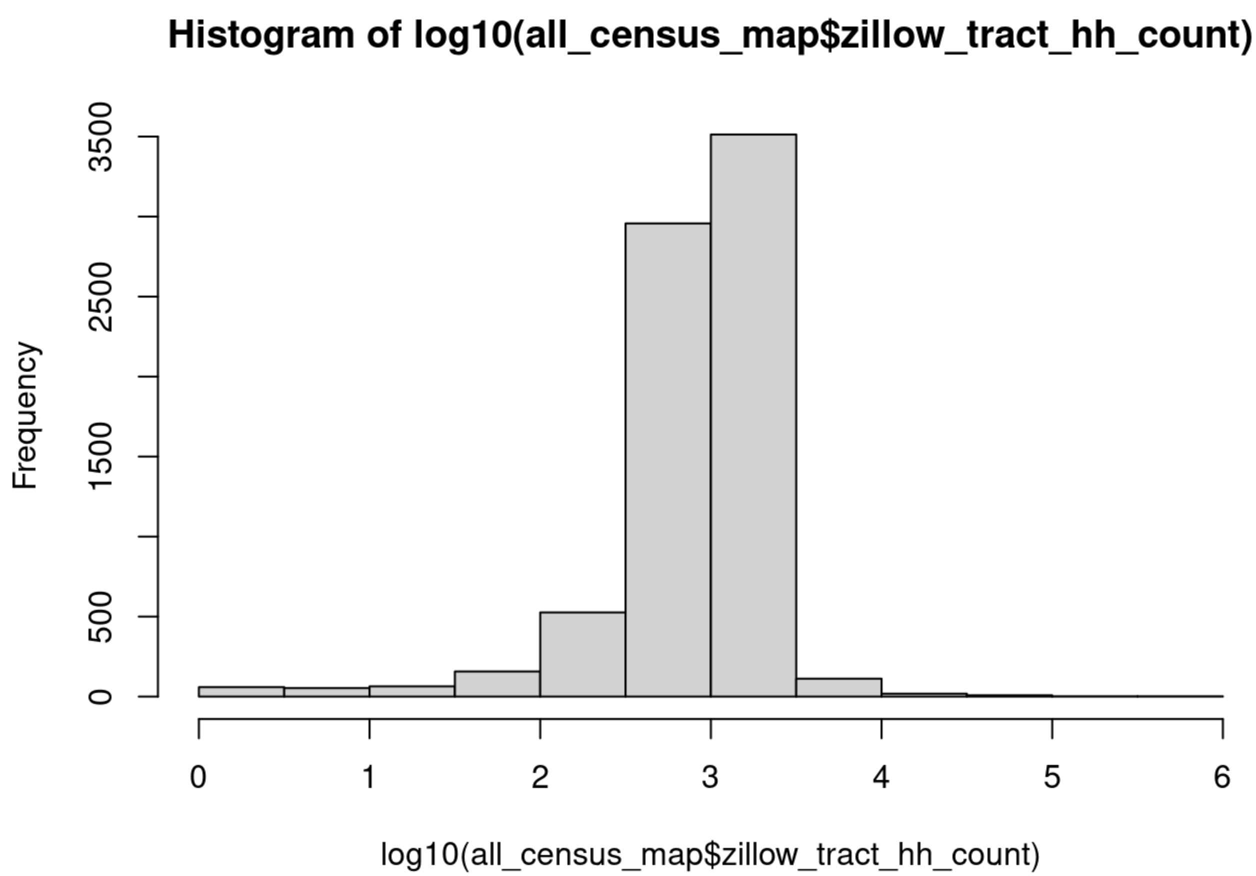
```
# iou service areas overlap, must handle overlapped tracts
overlap_tracts <- all_census_map |> # 1620 overlapped tracts
  group_by(geoid) |>
  filter(n() > 1) |>
  pull(geoid) |> unique()

# overlap_rows <- all_census_map |>
#   filter(geoid %in% overlap_tracts) |> arrange(geoid)

# fix overlapped tracts with weighted mean
all_census_map <- all_census_map |>
  group_by(geoid) |>
  summarise(
    across(starts_with("avg_der"), ~ weighted.mean(.x, w = zillow_tract_hh_count, na.rm =
    zillow_tract_hh_count = sum(zillow_tract_hh_count)
  ) |> ungroup()
```

```
# need to handle outliers in combined zillow estimate
all_census_map |>
  arrange(desc(zillow_tract_hh_count)) |>
  select(geoid, zillow_tract_hh_count) |>
  st_drop_geometry() |>
  head(20)
```

```
# A tibble: 20 × 2
   geoid       zillow_tract_hh_count
   <chr>                       <dbl>
 1 06085504805               426777
 2 06085504802               390285
 3 06073013317               160934
 4 06085505015               123319
 5 06085503127                65097
 6 06071002025                59387
 7 06057000104                51992
 8 06085504318                50609
 9 06077005137                47542
10 06037402403                41927
```

```
11 06085504506                32964
12 06085505014                31849
13 06065042523                31531
14 06095251802                31305
15 06085503304                29579
16 06095252904                19965
17 06081610303                19930.
18 06081608025                18889
19 06059062611                18004
20 06059063007                15530
```

```r
hist(log10(all_census_map$zillow_tract_hh_count))
```



The histogram shows our estimates are centered around 10^3 units, which, in the census data lines up with outliers above 10^4 should not be included.

```
all_census_map_filtered <- all_census_map |>
  filter(zillow_tract_hh_count <= 10000)
```

Random Forest Revised with only DAC data

```
features <- c(
  # Race/ethnicity (drop one to avoid perfect multicollinearity, drop White_Pct)
  "Hispanic_Pct", "Black_Pct", "Asian_Pct", "NatAmeri_Pct", "OtherMulti_Pct",

  # Socioeconomic
  "Poverty_Pctl", "HousingBurden_Pctl", "Education_Pctl", "CIscore_Pctl",
  "Unemployment_Pctl", "LinguisticIso_Pctl", "housing_density_hh_km2",

  # DAC status (binary)
  "is_dac"
)
```

```
# prepare clean model data
model_data <- all_census_map_filtered |>
  st_drop_geometry() |>
  left_join(dac |> st_drop_geometry(), by = join_by(geoid == tract)) |>
  mutate(
    is_dac = as.integer(CIscore_Pctl >= 75),
    diversity = 1 - ((White_Pct/100)^2 + (Hispanic_Pct/100)^2 + (Black_Pct/100)^2 +
                 (Asian_Pct/100)^2 + (NatAmeri_Pct/100)^2 + (OtherMulti_Pct/100)^2),
    housing_density_hh_km2 = zillow_tract_hh_count / Shape_Area * 1e6
  ) |>
  select(geoid, starts_with('avg_der'), all_of(features), zillow_tract_hh_count) |> na.omi
```

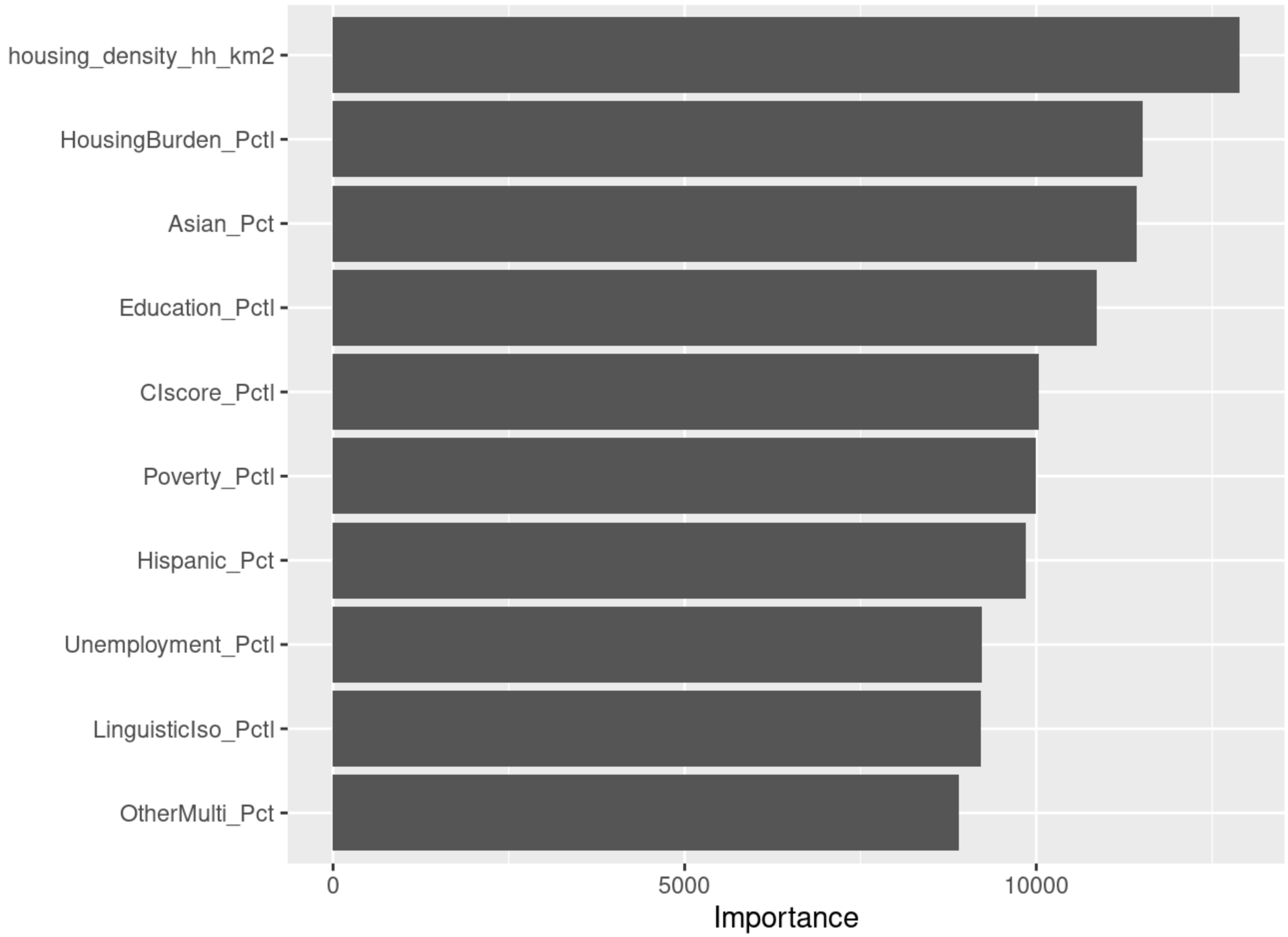
```
# 2. recipe
rec <- recipe(avg_der_remain_pv_hh ~ ., data = model_data |> select(geoid, avg_der_remain_
```

```r
  update_role(geoid, new_role = "ID") |>
  update_role(zillow_tract_hh_count, new_role = "ID")  # used as weight, not predictor

# 3. model spec — mtry to tune
rf_spec <- rand_forest(mtry = tune(), trees = 500) |>
  set_engine("ranger", importance = "impurity") |>
  set_mode("regression")

# 4. workflow
wf <- workflow() |>
  add_recipe(rec) |>
  add_model(rf_spec)

# 5. 10-fold CV + tune mtry
cv <- vfold_cv(model_data, v = 10)
tune_res <- tune_grid(
  wf,
  resamples = cv,
  grid = tibble(mtry = c(2, 3, 11))
)

# 6. fit final model with best mtry
best <- select_best(tune_res, metric = "rmse")
final_wf <- finalize_workflow(wf, best)
final_fit <- fit(final_wf, model_data)

# 7. results
collect_metrics(tune_res)
```

```
# A tibble: 6 × 7
   mtry .metric .estimator   mean     n std_err .config
  <dbl> <chr>   <chr>       <dbl> <int>   <dbl> <chr>
1     2 rmse    standard   4.07      10  0.115  pre0_mod1_post0
2     2 rsq     standard   0.103     10  0.0118 pre0_mod1_post0
3     3 rmse    standard   4.07      10  0.111  pre0_mod2_post0
4     3 rsq     standard   0.101     10  0.0119 pre0_mod2_post0
5    11 rmse    standard   4.13      10  0.102  pre0_mod3_post0
6    11 rsq     standard   0.0843    10  0.0109 pre0_mod3_post0
```

```
final_fit |> extract_fit_engine() |> vip::vip()
```

```
Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
ℹ Please use tidy evaluation idioms with `aes()`.
ℹ See also `vignette("ggplot2-in-packages")` for more information.
ℹ The deprecated feature was likely used in the vip package.
  Please report the issue at <https://github.com/koalaverse/vip/issues>.
```



The resulting metrics & VIP show that this RF model has a low R-squared but still has meaningful feature importance, indicating that our features don't *predict* hosting capacity but are still *correlated*.

```r
# post model checks
model_data |>
  select(all_of(features)) |>
  cor(use = "complete.obs") |>
  as.data.frame() |>
  rownames_to_column("var1") |>
  pivot_longer(-var1, names_to = "var2", values_to = "cor") |>
  filter(var1 < var2, abs(cor) > 0.5) |>
  arrange(desc(abs(cor)))
```

```
# A tibble: 14 × 3
   var1               var2                 cor
   <chr>              <chr>              <dbl>
 1 Education_Pctl     Hispanic_Pct       0.819
 2 CIscore_Pctl       Education_Pctl     0.805
 3 CIscore_Pctl       Poverty_Pctl       0.746
 4 CIscore_Pctl       Hispanic_Pct       0.730
 5 CIscore_Pctl       is_dac             0.727
 6 Education_Pctl     Poverty_Pctl       0.726
 7 Hispanic_Pct       Poverty_Pctl       0.618
 8 Education_Pctl     LinguisticIso_Pctl 0.611
 9 CIscore_Pctl       LinguisticIso_Pctl 0.605
10 Education_Pctl     is_dac             0.583
11 Hispanic_Pct       is_dac             0.572
12 HousingBurden_Pctl Poverty_Pctl       0.558
13 is_dac             Poverty_Pctl       0.533
14 Hispanic_Pct       OtherMulti_Pct    -0.508
```

As the data we used is from CalEnviroScreen and some variables are suspected to be composites of others, the above cell investigates our features for collinearity. Collinear variables will be dropped before running a linear regression. We found that `Education_Pctl` and `CIscore_Pctl` is correlated with `Hispanic_Pct`, `Poverty_Pctl`, and `LinguisticIso_Pctl`. `is_dac` will also be dropped as it does not show in our VIP.

```r
lm_features <- c(
  "housing_density_hh_km2", "Hispanic_Pct", "Black_Pct", "Asian_Pct", "NatAmeri_Pct",
```

```
  "OtherMulti_Pct", "Poverty_Pctl", "HousingBurden_Pctl", "Unemployment_Pctl", "Linguistic
)
```

```
# Linear Regression (scaled & weighted)
model_data_scaled <- model_data |>
  mutate(across(all_of(lm_features), scale))

lm_fit <- lm(avg_der_remain_pv_hh ~ ., data = model_data_scaled |> select(avg_der_remain_p
weights = model_data_scaled$zillow_tract_hh_count)

summary(lm_fit)
```

```
Call:
lm(formula = avg_der_remain_pv_hh ~ ., data = select(model_data_scaled,
    avg_der_remain_pv_hh, all_of(lm_features)), weights =
model_data_scaled$zillow_tract_hh_count)

Weighted Residuals:
    Min      1Q  Median      3Q     Max
-269.19  -49.03  -20.80   26.04 1048.29

Coefficients:
                        Estimate Std. Error t value Pr(>|t|)
(Intercept)             2.215288   0.035219  62.901  < 2e-16 ***
housing_density_hh_km2  0.002147   0.026569   0.081  0.93561
Hispanic_Pct            0.079997   0.061733   1.296  0.19507
Black_Pct              -0.039883   0.037782  -1.056  0.29118
Asian_Pct               0.134212   0.051721   2.595  0.00948 **
NatAmeri_Pct            0.011206   0.050409   0.222  0.82409
OtherMulti_Pct          0.066869   0.041409   1.615  0.10638
Poverty_Pctl           -0.547334   0.056640  -9.663  < 2e-16 ***
HousingBurden_Pctl      0.512746   0.043896  11.681  < 2e-16 ***
Unemployment_Pctl      -0.090281   0.038640  -2.336  0.01949 *
LinguisticIso_Pctl      0.091622   0.056189   1.631  0.10302
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

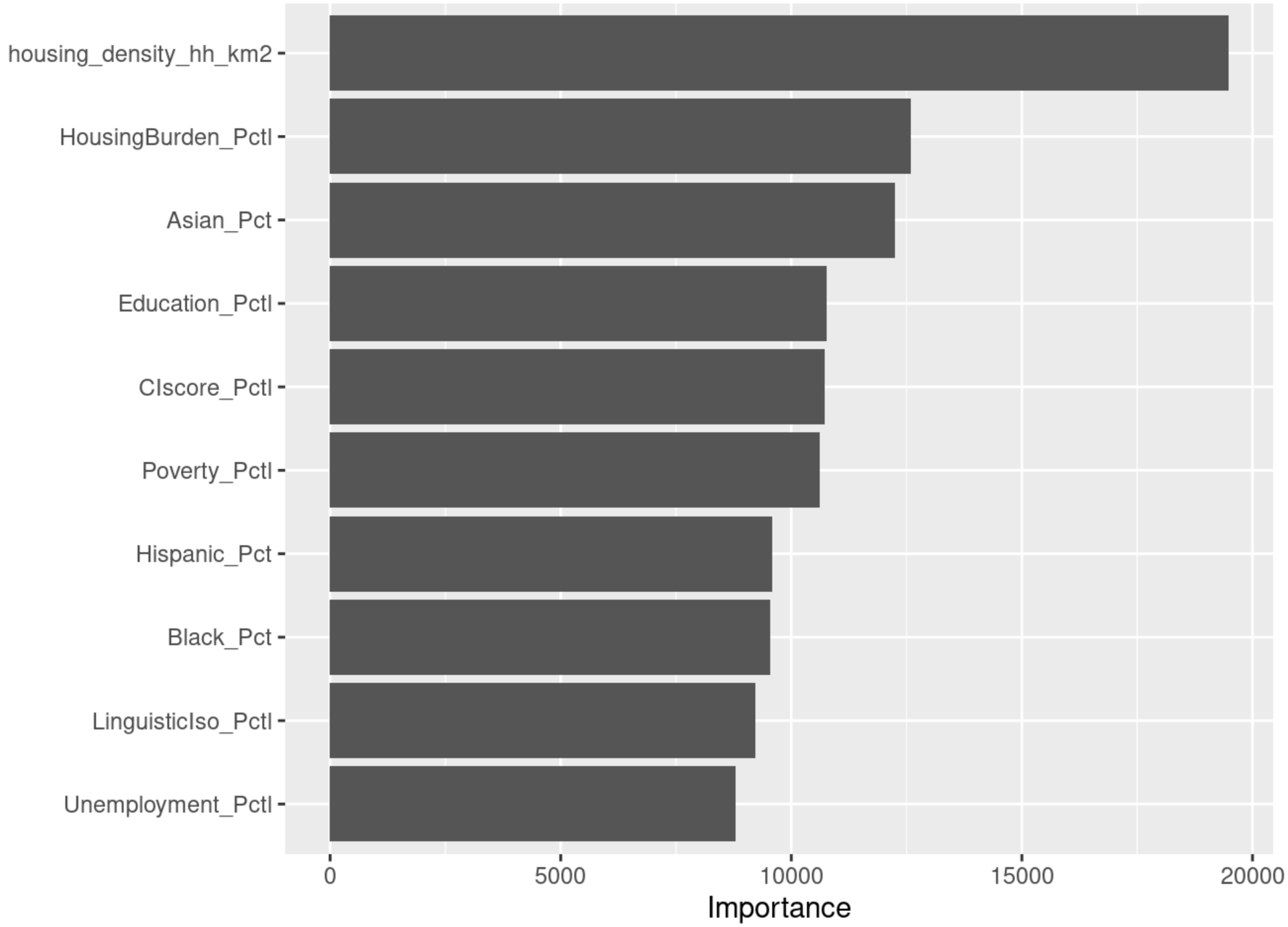
```
Residual standard error: 98.11 on 7350 degrees of freedom
Multiple R-squared:  0.03307,   Adjusted R-squared:  0.03175
F-statistic: 25.14 on 10 and 7350 DF,  p-value: < 2.2e-16
```

Before further investigation, we repeat this same process on another DER type.

```
# 2. recipe
rec <- recipe(avg_der_remain_load_hh ~ ., data = model_data |> select(geoid, avg_der_remai
  update_role(geoid, new_role = "ID") |>
  update_role(zillow_tract_hh_count, new_role = "ID")  # used as weight, not predictor

# 3. model spec — mtry to tune
rf_spec <- rand_forest(mtry = tune(), trees = 500) |>
  set_engine("ranger", importance = "impurity") |>
  set_mode("regression")

# 4. workflow
wf <- workflow() |>
  add_recipe(rec) |>
  add_model(rf_spec)

# 5. 10-fold CV + tune mtry
cv <- vfold_cv(model_data, v = 10)
tune_res <- tune_grid(
  wf,
  resamples = cv,
  grid = tibble(mtry = c(2, 3, 11))
)

# 6. fit final model with best mtry
best <- select_best(tune_res, metric = "rmse")
final_wf <- finalize_workflow(wf, best)
final_fit <- fit(final_wf, model_data)

# 7. results
collect_metrics(tune_res)
```

```
# A tibble: 6 × 7
   mtry .metric .estimator   mean     n std_err .config
  <dbl> <chr>   <chr>       <dbl> <int>   <dbl> <chr>
1     2 rmse    standard   4.17      10 0.131   pre0_mod1_post0
2     2 rsq     standard   0.0965    10 0.00851 pre0_mod1_post0
3     3 rmse    standard   4.17      10 0.130   pre0_mod2_post0
4     3 rsq     standard   0.100     10 0.0107  pre0_mod2_post0
5    11 rmse    standard   4.23      10 0.123   pre0_mod3_post0
6    11 rsq     standard   0.0978    10 0.0143  pre0_mod3_post0
```

```
final_fit |> extract_fit_engine() |> vip::vip()
```



```
lm_fit <- lm(avg_der_remain_load_hh ~ ., data = model_data_scaled |> select(avg_der_remain
weights = model_data_scaled$zillow_tract_hh_count)

summary(lm_fit)
```

```
Call:
lm(formula = avg_der_remain_load_hh ~ ., data = select(model_data_scaled,
    avg_der_remain_load_hh, all_of(lm_features)), weights =
model_data_scaled$zillow_tract_hh_count)

Weighted Residuals:
    Min      1Q  Median      3Q     Max
-155.72  -37.71  -15.90   21.10 1062.26

Coefficients:
                        Estimate Std. Error t value Pr(>|t|)
(Intercept)             1.815951   0.029414  61.739   <2e-16 ***
housing_density_hh_km2 -0.206400   0.022190  -9.302   <2e-16 ***
Hispanic_Pct           -0.120710   0.051558  -2.341   0.0192 *
Black_Pct              -0.050555   0.031554  -1.602   0.1092
Asian_Pct               0.056191   0.043196   1.301   0.1934
NatAmeri_Pct           -0.005161   0.042100  -0.123   0.9024
OtherMulti_Pct          0.050099   0.034583   1.449   0.1475
Poverty_Pctl            0.077494   0.047304   1.638   0.1014
HousingBurden_Pctl     -0.067253   0.036661  -1.834   0.0666 .
Unemployment_Pctl      -0.010619   0.032271  -0.329   0.7421
LinguisticIso_Pctl      0.015637   0.046927   0.333   0.7390
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 81.94 on 7350 degrees of freedom
Multiple R-squared:  0.01659,   Adjusted R-squared:  0.01525
F-statistic:  12.4 on 10 and 7350 DF,  p-value: < 2.2e-16
```

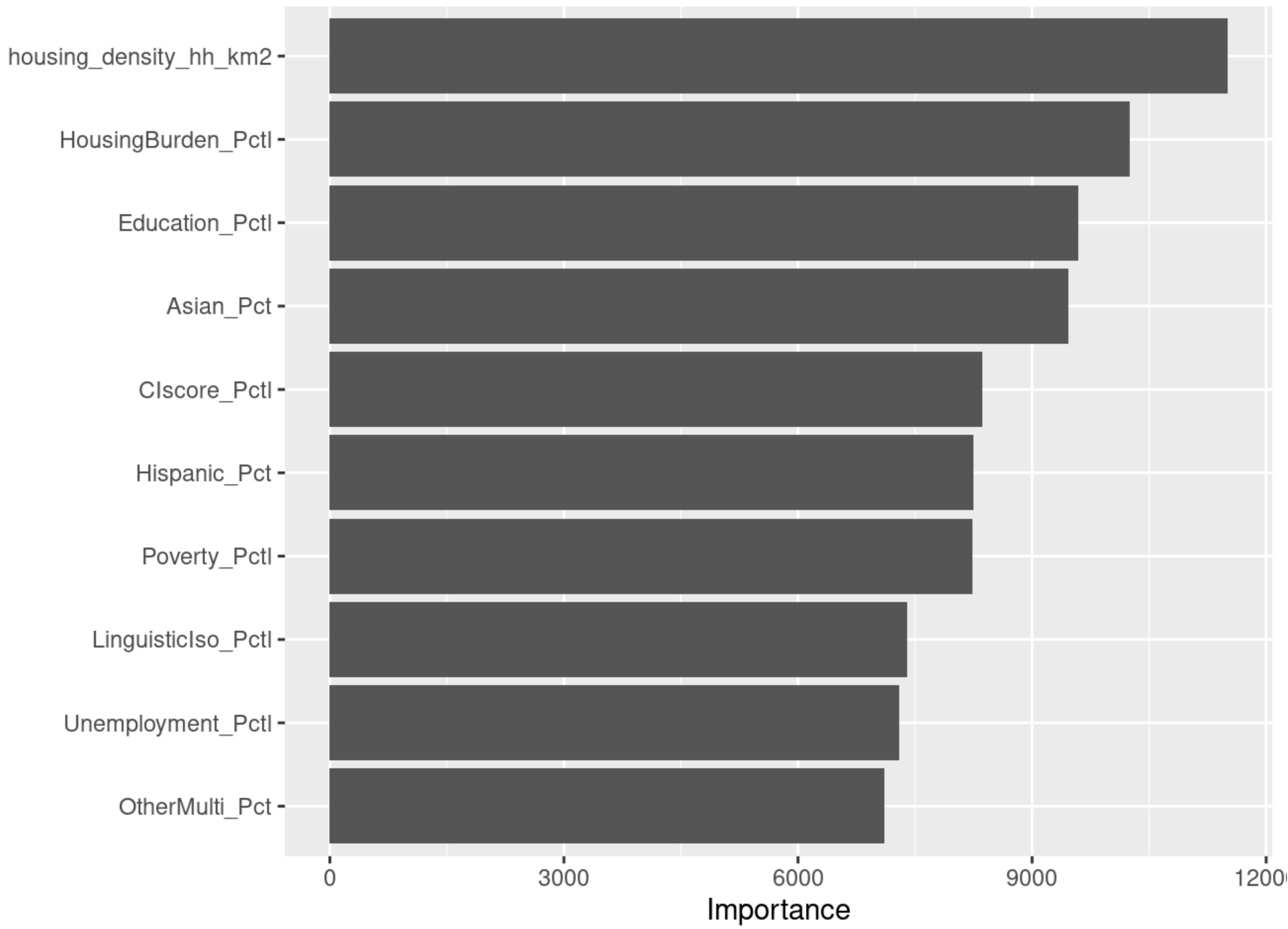
Run our models again on remaining generation.

```r
# 2. recipe
rec <- recipe(avg_der_remain_generation_hh ~ ., data = model_data |> select(geoid, avg_der
  update_role(geoid, new_role = "ID") |>
  update_role(zillow_tract_hh_count, new_role = "ID")  # used as weight, not predictor

# 3. model spec — mtry to tune
rf_spec <- rand_forest(mtry = tune(), trees = 500) |>
  set_engine("ranger", importance = "impurity") |>
  set_mode("regression")

# 4. workflow
wf <- workflow() |>
  add_recipe(rec) |>
  add_model(rf_spec)

# 5. 10-fold CV + tune mtry
cv <- vfold_cv(model_data, v = 10)
tune_res <- tune_grid(
  wf,
  resamples = cv,
  grid = tibble(mtry = c(2, 3, 11))
)

# 6. fit final model with best mtry
best <- select_best(tune_res, metric = "rmse")
final_wf <- finalize_workflow(wf, best)
final_fit <- fit(final_wf, model_data)

# 7. results
collect_metrics(tune_res)
```

```
# A tibble: 6 × 7
   mtry .metric .estimator   mean     n std_err .config
  <dbl> <chr>   <chr>       <dbl> <int>   <dbl> <chr>
1     2 rmse    standard   3.66      10  0.167  pre0_mod1_post0
2     2 rsq     standard   0.0922    10  0.0121 pre0_mod1_post0
3     3 rmse    standard   3.66      10  0.164  pre0_mod2_post0
```

```
4      3 rsq     standard   0.0933    10  0.0135 pre0_mod2_post0
5     11 rmse    standard   3.71      10  0.160  pre0_mod3_post0
6     11 rsq     standard   0.0799    10  0.0143 pre0_mod3_post0
```

```
final_fit |> extract_fit_engine() |> vip::vip()
```




```
lm_fit <- lm(avg_der_remain_generation_hh ~ ., data = model_data_scaled |> select(avg_der_
weights = model_data_scaled$zillow_tract_hh_count)

summary(lm_fit)
```

```
Call:
lm(formula = avg_der_remain_generation_hh ~ ., data = select(model_data_scaled,
    avg_der_remain_generation_hh, all_of(lm_features)), weights =
model_data_scaled$zillow_tract_hh_count)

Weighted Residuals:
    Min      1Q  Median      3Q     Max
-223.85  -41.75  -18.10   20.61 1023.59

Coefficients:
                       Estimate Std. Error t value Pr(>|t|)
(Intercept)             1.78352    0.03127  57.041   <2e-16 ***
housing_density_hh_km2  0.01661    0.02359   0.704   0.4812
Hispanic_Pct            0.03014    0.05481   0.550   0.5824
Black_Pct              -0.02393    0.03354  -0.713   0.4756
Asian_Pct               0.11537    0.04592   2.512   0.0120 *
NatAmeri_Pct           -0.02564    0.04475  -0.573   0.5667
OtherMulti_Pct          0.08196    0.03676   2.229   0.0258 *
Poverty_Pctl           -0.42886    0.05029  -8.529   <2e-16 ***
HousingBurden_Pctl      0.42342    0.03897  10.865   <2e-16 ***
Unemployment_Pctl      -0.07711    0.03431  -2.248   0.0246 *
LinguisticIso_Pctl      0.08595    0.04988   1.723   0.0849 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 87.1 on 7350 degrees of freedom
Multiple R-squared:  0.03062,   Adjusted R-squared:  0.0293
F-statistic: 23.22 on 10 and 7350 DF,  p-value: < 2.2e-16
```

———–– everything below is for previous iterations of a potential stats workflow, keeping it for now ——–

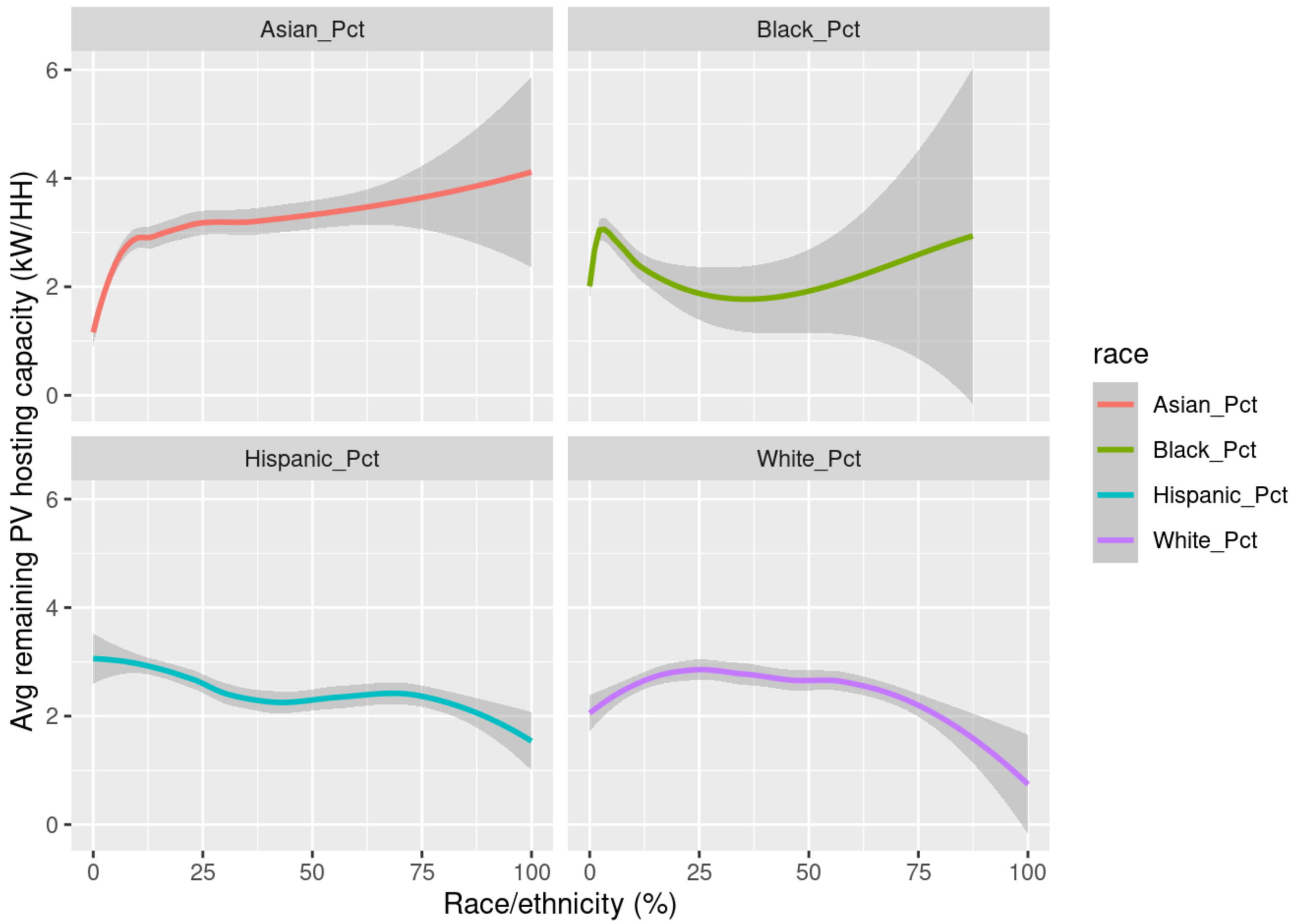
Prepare data for statistics

```
# calenviroscreen join
der_cal <- left_join(all_census_map, dac %>% st_drop_geometry(), by = join_by(geoid == tra
```

```
der_cal |>
  select(geoid, avg_der_remain_pv_hh, White_Pct, Hispanic_Pct, Black_Pct, Asian_Pct) |>
  pivot_longer(cols = ends_with('_Pct'), names_to = 'race', values_to = 'race_pct') |>
  ggplot(aes(x = race_pct, y = avg_der_remain_pv_hh, color = race)) +
  geom_smooth(method = 'loess', se = TRUE) +
  facet_wrap(~race) +
  labs(x = 'Race/ethnicity (%)', y = 'Avg remaining PV hosting capacity (kW/HH)')
```

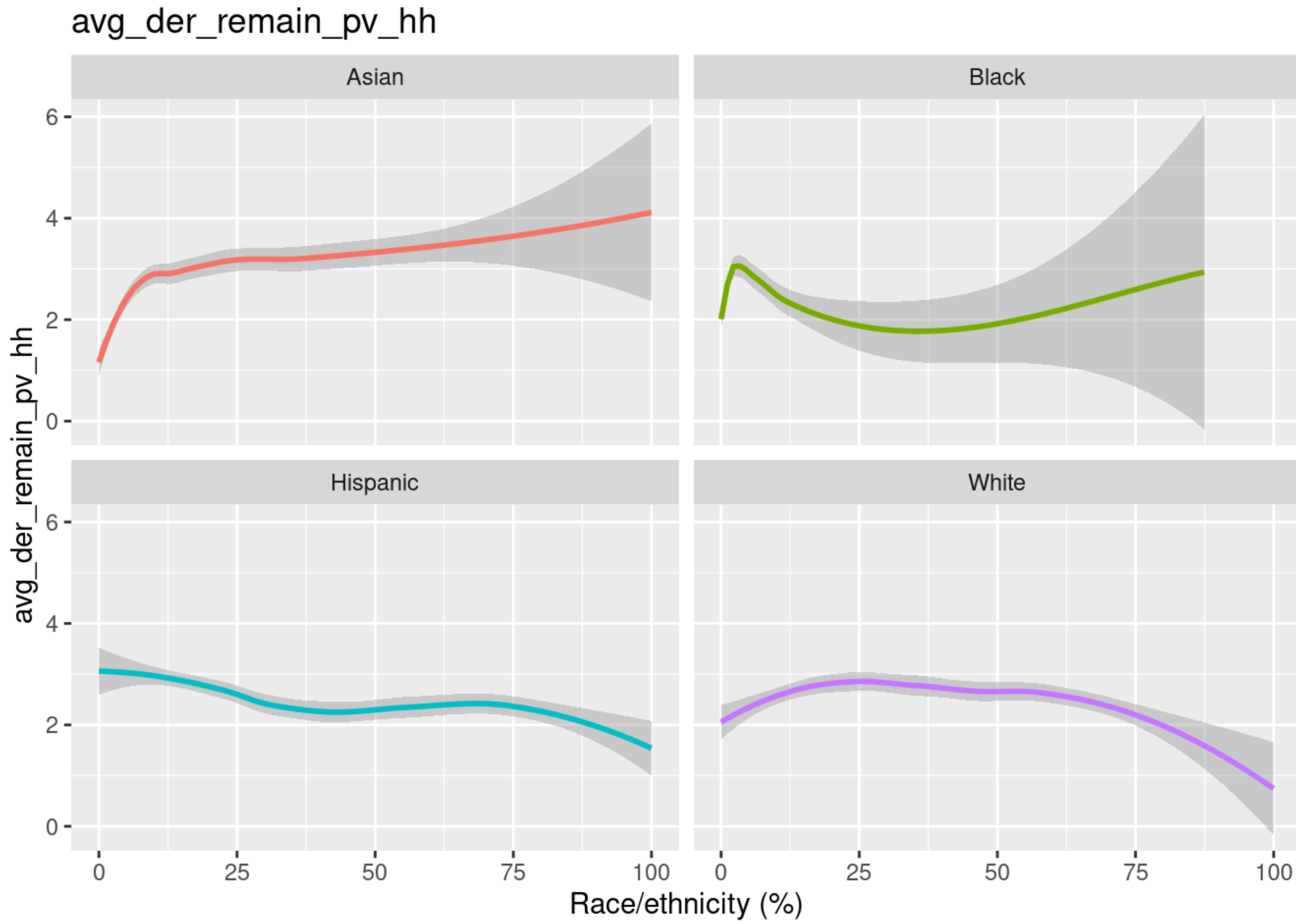
```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 52 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

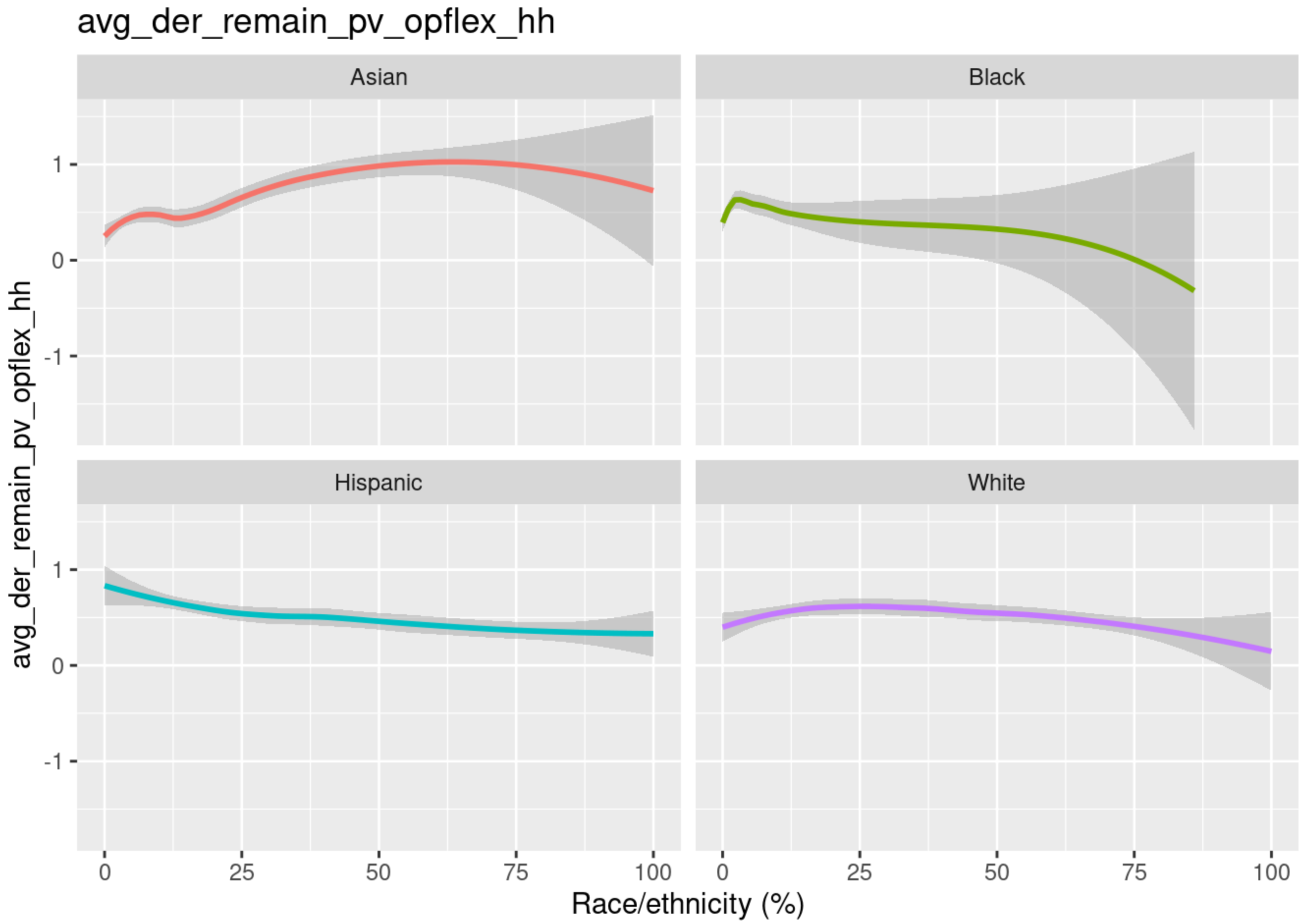


```
der_cols <- der_cal |> st_drop_geometry() |> select(starts_with('avg_der')) |> names()

plots <- map(der_cols, \(der_type) {
  der_cal |>
    select(geoid, all_of(der_type), White_Pct, Hispanic_Pct, Black_Pct, Asian_Pct) |>
    pivot_longer(cols = ends_with('_Pct'), names_to = 'race', values_to = 'race_pct') |>
    mutate(race = str_remove(race, '_Pct')) |>
    ggplot(aes(x = race_pct, y = .data[[der_type]], color = race)) +
    geom_smooth(method = 'loess', se = TRUE) +
    guides(color = 'none') +
    facet_wrap(~race) +
    labs(x = 'Race/ethnicity (%)', y = der_type, title = der_type)
})

walk(plots, print)
```

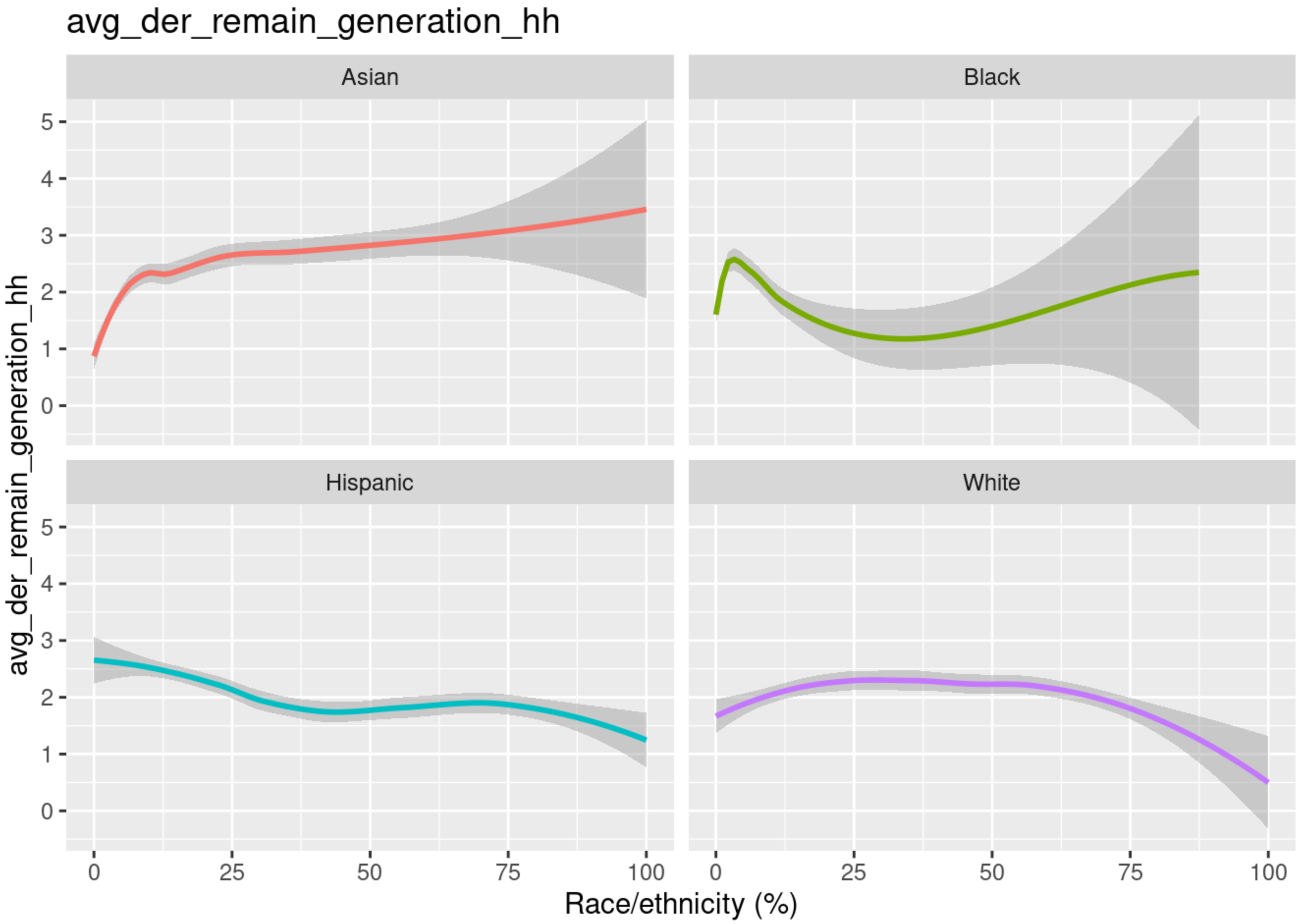
```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 52 rows containing non-finite outside the scale range
(`stat_smooth()`).
```



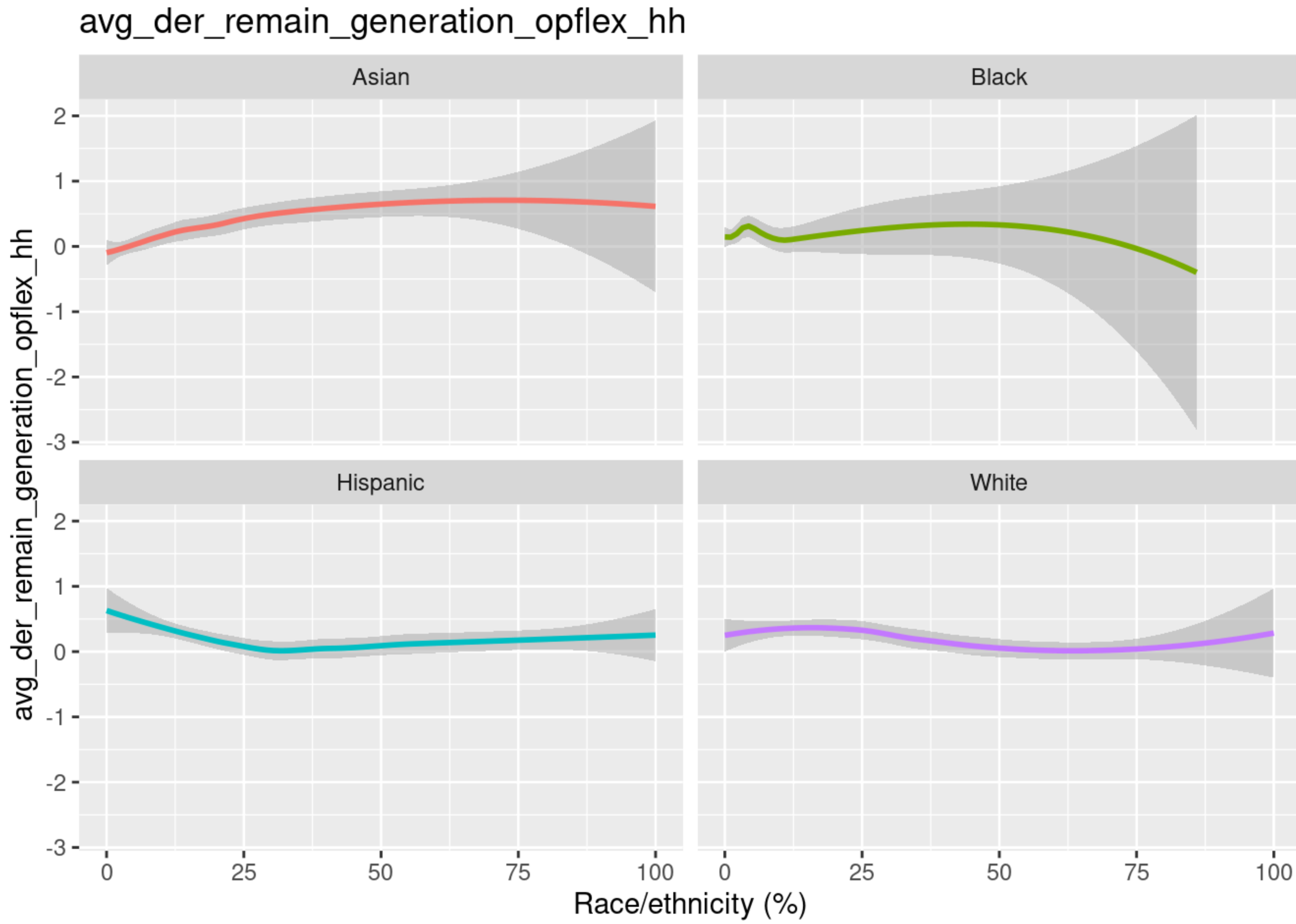

```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 284 rows containing non-finite outside the scale range
(`stat_smooth()`).
```



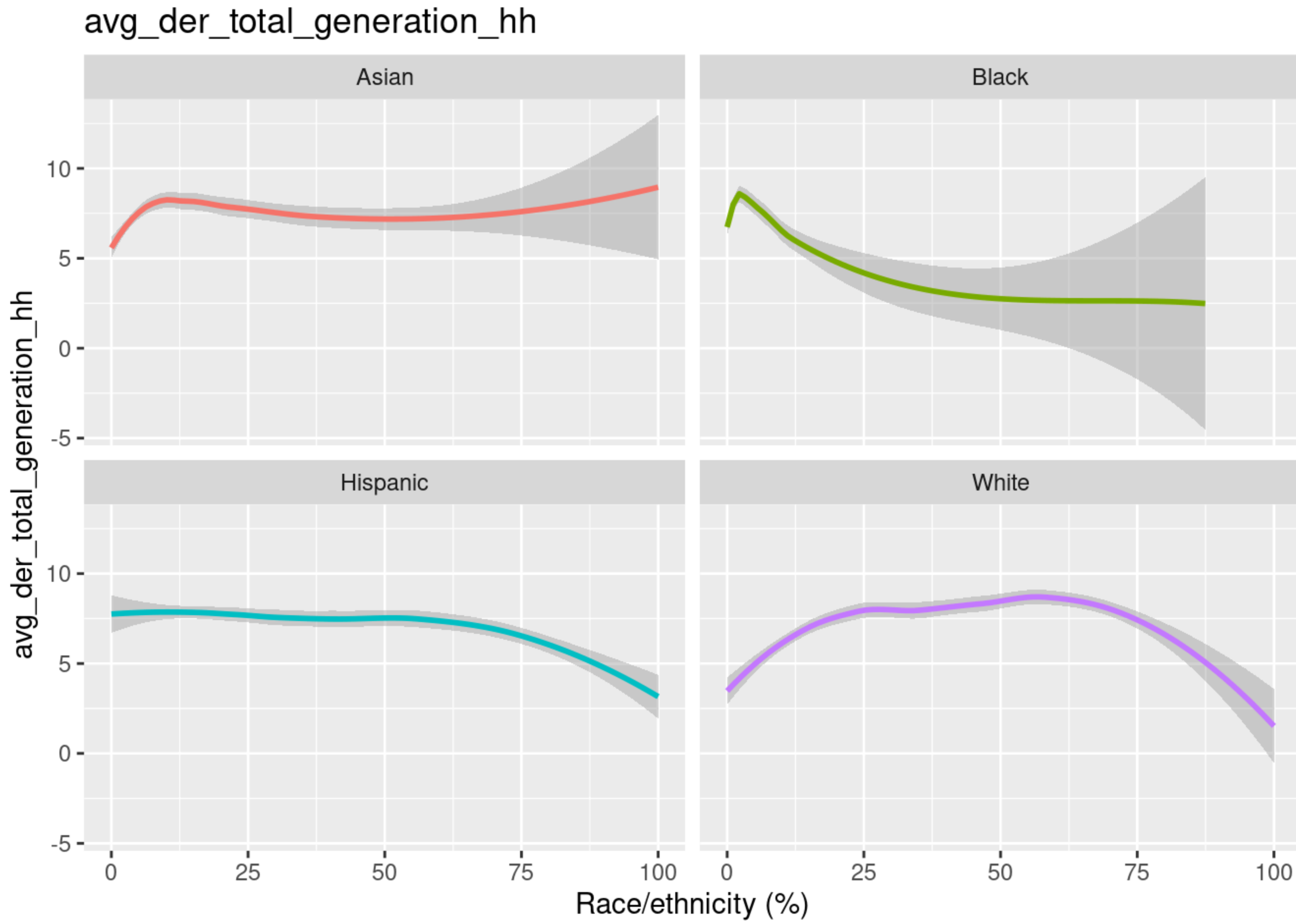

```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 48 rows containing non-finite outside the scale range
(`stat_smooth()`).
```



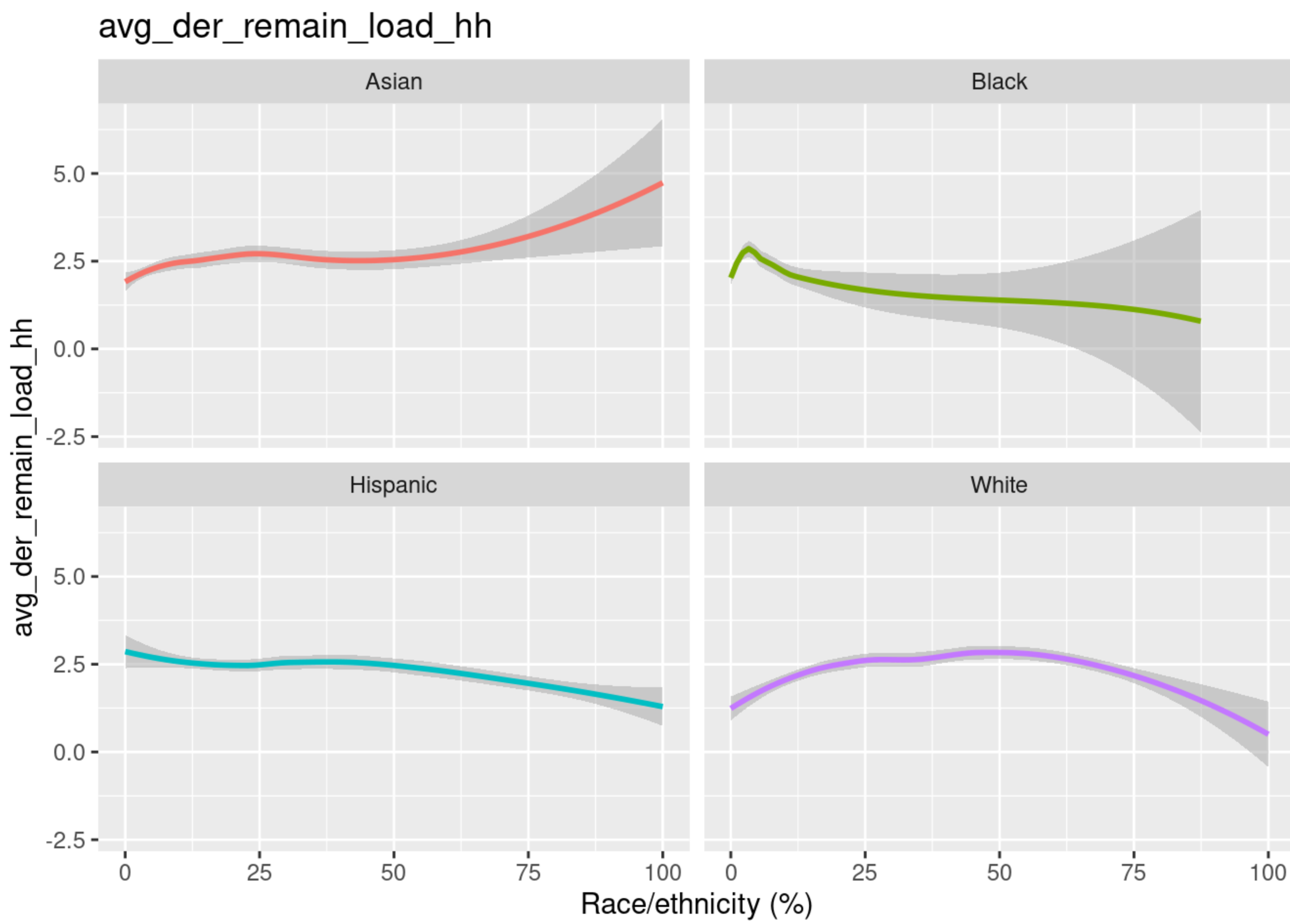

```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 280 rows containing non-finite outside the scale range
(`stat_smooth()`).
```



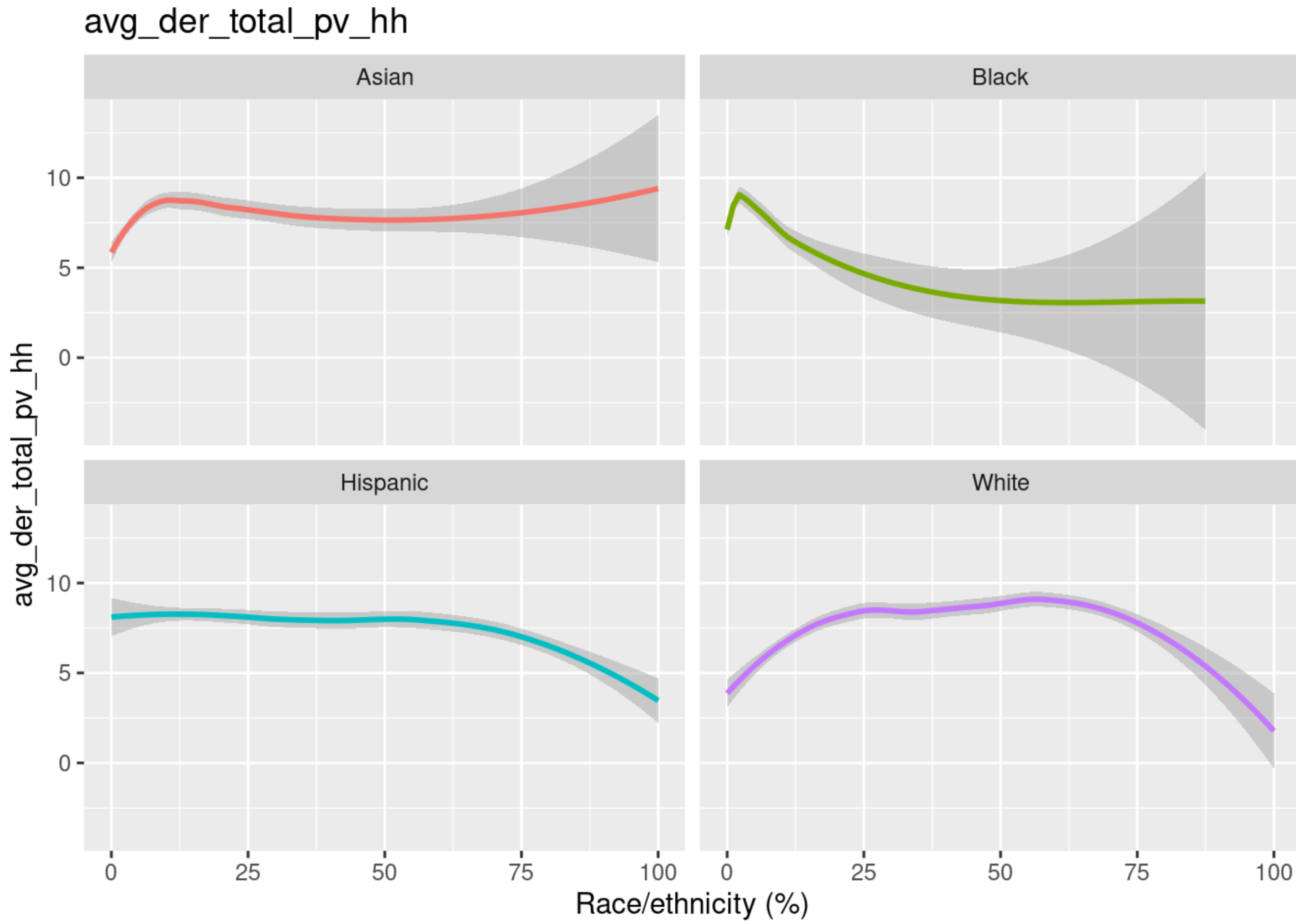

```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 56 rows containing non-finite outside the scale range
(`stat_smooth()`).
```




```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 48 rows containing non-finite outside the scale range
(`stat_smooth()`).
```




```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 60 rows containing non-finite outside the scale range
(`stat_smooth()`).
```




```
model <- lm(avg_der_total_pv_hh ~ White_Pct + Hispanic_Pct + Asian_Pct + Black_Pct, data =
summary(model)
```

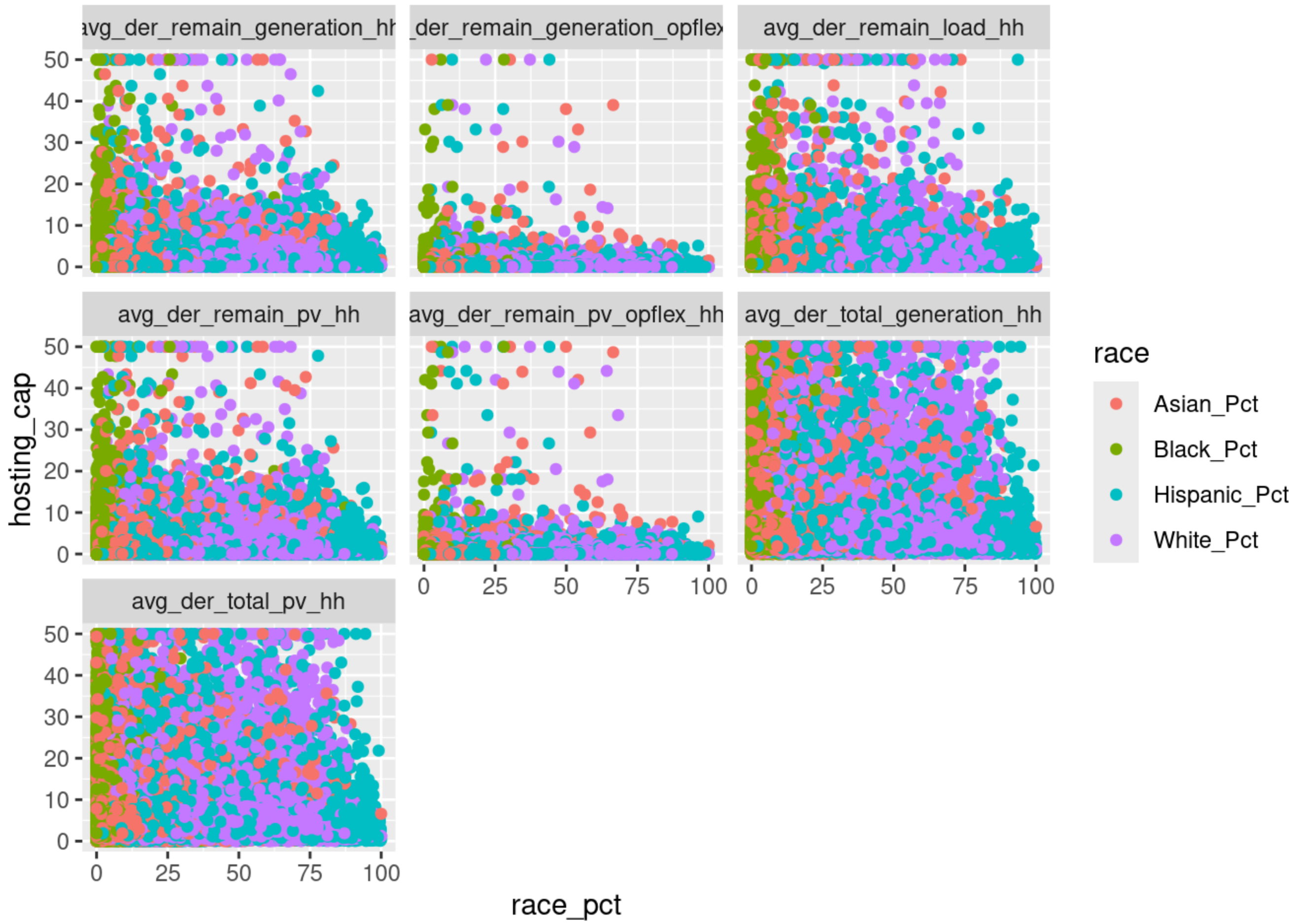
```
Call:
lm(formula = avg_der_total_pv_hh ~ White_Pct + Hispanic_Pct +
    Asian_Pct + Black_Pct, data = der_cal)

Residuals:
   Min      1Q Median      3Q     Max
-9.233 -5.617 -3.537  0.647 44.570
```

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  11.06641    3.29901   3.354 0.000799 ***
White_Pct    -0.01927    0.03598  -0.536 0.592269
Hispanic_Pct -0.04505    0.03339  -1.349 0.177340
Asian_Pct    -0.02344    0.03522  -0.666 0.505731
Black_Pct    -0.10290    0.03981  -2.585 0.009753 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.02 on 7451 degrees of freedom
  (15 observations deleted due to missingness)
Multiple R-squared:  0.008919,  Adjusted R-squared:  0.008387
F-statistic: 16.76 on 4 and 7451 DF,  p-value: 1.094e-13
```

```
der_long <- der_cal %>%
  select(geoid, starts_with('avg_der'), 'White_Pct', 'Hispanic_Pct', 'Black_Pct', 'Asian_P
  pivot_longer(cols=starts_with('avg_der'), names_to = 'der_type', values_to = 'hosting_ca
  pivot_longer(cols=ends_with('Pct'), names_to = 'race', values_to = 'race_pct')
```

```
ggplot(der_long %>% filter(hosting_cap >= 0), aes(x = race_pct, y = hosting_cap, color = r
  geom_point() +
  facet_wrap(~der_type)
```

```
Warning: Removed 124 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
# acs join
joined <- left_join(all_census_map, census, by = 'geoid')

joined_cleandem <- joined |>
    filter(str_starts(variable, "B01003_") | str_starts(variable, "B02001")) |>
    mutate(demographic = case_when(
        str_detect(label, "White") ~ "White",
        str_detect(label, "Black") ~ "Black",
        str_detect(label, "Total:!!Hispanic") ~ "Hispanic/Latino",
        str_detect(label, "Indian") ~ "American Indian/Alaska Native",
        str_detect(label, "Asian") ~ "Asian",
```

```
        str_detect(label, "Hawaiian") ~ "Native Hawaiian/Pacific Islander",
        str_detect(label, "Other Race") ~ "Other Race",
        str_detect(label, "Two or More") ~ "2+ Races",
        str_detect(label, "Total:!!Not") ~ "Not Hispanic/Latino",
        str_detect(label, "Estimate!!Total") ~ "Total Pop"
    )) |>
    relocate(demographic, .after = label)

joined_wide <- joined_cleandem |>
  sf::st_drop_geometry() |>
  pivot_wider(
    id_cols = c(geoid, starts_with('avg_der'), name),
    names_from = demographic,
    values_from = estimate) |>
  janitor::clean_names() |>
  mutate(
    diversity = 1 - (white*(white-1) + black*(black-1) + american_indian_alaska_native*(am

percents <- joined_wide |>
  mutate(
    white_pct = white / total_pop,
    black_pct = black / total_pop,
    native_indian_pct = american_indian_alaska_native / total_pop,
    asian_pct = asian / total_pop,
    native_pacific_pct = native_hawaiian_pacific_islander / total_pop,
    other_pct = other_race / total_pop,
    x2_races_pct = x2_races / total_pop) |>
      select(geoid, name, starts_with('avg_der'), ends_with('_pct'), diversity)
```

```
model <- lm(avg_der_remain_pv_hh ~ white_pct + black_pct + asian_pct + x2_races_pct + dive
summary(model)
```

```
Call:
lm(formula = avg_der_remain_pv_hh ~ white_pct + black_pct + asian_pct + 
    x2_races_pct + diversity, data = percents)
```

```
Residuals:
   Min     1Q Median     3Q    Max 
-3.795 -2.090 -1.177  0.619 47.865 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)   0.35279    0.49190   0.717   0.4733    
white_pct     1.18035    0.39899   2.958   0.0031 ** 
black_pct    -0.05381    0.74782  -0.072   0.9426    
asian_pct     3.09184    0.41069   7.528 5.74e-14 ***
x2_races_pct -1.14060    0.80212  -1.422   0.1551    
diversity     2.32197    0.54220   4.282 1.87e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.318 on 7452 degrees of freedom
  (13 observations deleted due to missingness)
Multiple R-squared:  0.01549,	Adjusted R-squared:  0.01483 
F-statistic: 23.45 on 5 and 7452 DF,  p-value: < 2.2e-16
```